WIDIANA ANGGREANI RAHWAL

https://widiaangrn.netlify.app | widia.angrn@gmail.com | www.linkedin.com/in/widiaangrn

PROFESSIONAL SUMMARY

Fullstack developer with experience building an oil and gas well monitoring system for Pertamina Hulu Mahakam using ASP.NET MVC and Oracle SQL, as well as web and mobile applications for the healthcare, education, tourism, and retail sectors using Django and Flutter. Capable of handling the full development lifecycle: requirements analysis, UI/UX design, frontend and backend development, database optimization, and payment integration.

SKILLS

- **Programming Languages:** Python, Dart, JavaScript (jQuery), PHP, C#
- **Frameworks:** Django, Flutter, Laravel, ASP.NET MVC, React.js (Basic)
- **Databases:** PostgreSQL, MySQL, Oracle SQL, MsSQL
- **Development Tools:** Git, Postman, Docker (Basic), VirtualBox, XAMPP, Laragon
- **Design & Collaboration Tools:** Figma, Draw.io, Miro, Trello
- **Front-End Tools:** Bootstrap, HTML, CSS, Chart.js, ECharts
- **Non-Technical**: Analytical thinking, time management, teamwork & collaboration, adaptability, problem solving.
- **Languages**: Indonesian (Native), English (Intermediate | IAET ITK: 520, TOEFL Prediction: 610)

WORK EXPERIENCE

Programmer **January 2024 – September 2026**
PT. Sila Teknologi Utama

Key Achievements:

- Developed 19 functional requirements (9 dashboards & toolboxes, 10 other features) for the Pertamina Hulu Mahakam oil and gas well monitoring system using ASP.NET MVC and MSSQL / Oracle SQL.
- Revamped application performance by migrating business logic and calculations from the application layer to Stored Procedures and SQL Views with parameters and pagination, reducing the data transferred per request from millions of records to 50 records — significantly improving access speed.
- Contributed to the development of more than 10 cross-sector applications (education, healthcare, retail, and tourism) using Django and Flutter backed by PostgreSQL database.
- Automated the oil and gas well monitoring system, replacing manual Excel-based processes and speeding up data analysis.
- Developed a ticketing platform, a multi-method payment system (OTS, QRIS, EDC) integrated with Midtrans, and a voucher and promotion system for the tourism and retail sectors.
- Optimized the backend architecture and system security to improve scalability and code maintainability.

Project Details:

A. Industrial Automation & Monitoring System Development (Client: Pertamina Hulu Mahakam)

- Designed and developed interactive custom charts using Chart.js and Echarts to visualize gas, oil, and sand risk assessment data, enabling faster problem identification.
- Designed an Entity-Relationship Diagram (ERD) to consolidate multiple databases into a centralized data warehouse, improving data structure and ease of control.
- Developed 19 functional requirements using ASP.NET MVC and MSSQL / Oracle SQL, covering 9 dashboards / toolboxes and 10 supporting features such as data tables, Excel upload, and a landing page.
- Performed application performance revamping by moving calculation logic from the application layer to Stored Procedures and SQL Views.

B. Website Feature Development (Django)

- Improved the authentication system by applying OOP principles, Role-Based Access Control (RBAC), and automatic activation token delivery via email to enhance user registration security.
- Helped develop a real-time chat feature using Django Channels, ASGI, and Redis to support stable and responsive two-way communication.
- Helped develop a complete payment module (OTS, QRIS, EDC, etc.) with Midtrans integration, along with a voucher and promo system that contributed to increased transactions.
- Developed a scheduler with django-crontab to automate daily data synchronization and delivery of notification emails (alarms, transaction confirmations).
- Redesigned several pages and developed several other key features.
- Developed a patient registration module by integrating a queue system, barcodes, and online forms, cutting registration time and minimizing input errors.

C. Mobile Application Development & Integration (Flutter)

- Designed and created the application UI/UX.
- Developed Flutter applications and REST APIs to ensure reliable real-time data synchronization for transactions, stock, prices, tickets, and other data.

- Developed medical sample identification by building a barcode-based labeling system that can be scanned to instantly display patient data.

D. E-Learning & Assessment System Development

- Developed a multi-user Exam System from UI/UX design to implementation, covering 4 user roles (Admin, Teacher, Exam Proctor, Student) with different access rights and workflows.
- Built a real-time exam proctoring module, covering student activity monitoring, lock/unlock access, and remaining time display.
- Configured Safe Exam Browser (SEB) to restrict laptop exam access to a controlled browser environment.

Software Quality Assurance **October 2023 – December 2023**

PT.Mahakarya Inovasi Bangsa

- Conducted User Acceptance Testing (UAT) and System Integration Testing (SIT) on a Laravel website and prepared testing result documentation.
- Analyzed bugs, reported findings to the developer team, and produced comprehensive test reports with Trello.

EDUCATION & TRAINING

Institut Teknologi Kalimantan **2019 – 2023**

Bachelor of Informatics, 3.76/4.00

- Built the ITK Internal Quality Assurance Information System end-to-end by applying the Personal Extreme Programming method, using Laravel and MySQL to ease academic accreditation data collection and reporting.
- Built various learning projects to master full-stack development using Laravel (backend), React.js (frontend), and MySQL (database).
- Technical: PHP, Laravel, MySQL, React.js, VirtualBox, Docker (Basic).

IDCamp 2024 **September 2024 – May 2025**

Multi-Platform App Developer

- Strong foundation in Object-Oriented Programming (OOP) and a commitment to writing clean code through the application of best practices and effective debugging techniques.
- Skilled in application development using Flutter, covering responsive UI/UX design, implementation of state management with Provider, and reliable REST API integration.

Google Bangkit 2022 **February 2022 – July 2022**

Android Developer Learning Path

- Developed DentsHealth, a dental health diagnostic app built with Kotlin that uses Natural Language Processing (NLP) to analyze user complaints and cloud computing to deploy a Machine Learning model, enabling automated detection of dental and oral problems.
- Learned hard skills for building Android applications, including the Kotlin programming language, Android app fundamentals, UX design, SOLID programming principles, and the OOP paradigm.
- Learned soft skills needed in the workplace such as time management, adaptability, and critical thinking, as well as spoken and written English proficiency.

ORGANIZATIONAL EXPERIENCE

Informatics Student Association **September 2020 – February 2021**

Information Media Staff

- Designed HMIF Instagram content using Adobe Illustrator and CorelDraw and participated as a committee member in organizational events.

ACHIEVEMENTS & OTHER EXPERIENCE

- **Copyright & Certifications**: Copyright for PsyCare software, 15+ technology certifications from Dicoding, IDCamp, Google Bangkit, and other programs including Flutter, Android Development, Dart, SOLID Principles, and Fundamentals of AI.
- **Other Projects**: CISA (Covid-19 Information and Self Check Up Application), PetYourself, PsyCare, SIKMA (Internship Information System), DentsHealth, Restaurant App, SIMJAMU (Quality Assurance System), LeafPad, Gallery App.
- **Volunteer Work**: Mangrove Warrior 2025, Lestravida Balikpapan Batch 1.